

GIBRALTAR AP, GibraltarWMO: **084950**Lat: **36.1521N**Lon: **5.3494W**Elev: **15**StdP: **14.69**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	46.4	48.4	28.6	22.6	54.3	33.5	28.1	54.1	40.4	57.1	35.0	57.2	8.2	270	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	87.5	68.5	84.4	68.1	82.3	68.2	74.3	78.7	73.3	77.9	72.4	77.2	13.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.8	121.9	76.8	71.8	117.6	76.1	70.9	113.8	75.5	37.7	78.8	36.9	77.9	36.1	77.1	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.6	27.3	24.0	DB	42.2	93.8	3.0	3.6	40.1	96.4	38.4	98.6	36.7	100.6	34.5	103.2
			WB	36.5	76.7	3.1	2.1	34.3	78.2	32.5	79.4	30.7	80.6	28.5	82.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.6	56.7	57.3	59.8	62.4	66.5	71.5	75.5	76.3	72.8	68.1	61.8	58.4	(6)
(7)		DBStd	7.60	3.28	3.29	3.04	3.02	3.33	3.10	3.28	2.89	3.23	3.26	3.94	3.23	(7)
(8)		HDD50	5	3	1	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD65	1073	258	216	164	88	21	0	0	0	0	10	110	206	(9)
(10)		CDD50	5708	209	205	303	371	511	646	789	815	684	561	354	260	(10)
(11)		CDD65	1301	0	0	2	9	67	196	324	350	234	106	13	0	(11)
(12)		CDH74	5525	0	0	6	12	134	645	1820	2035	744	120	9	0	(12)
(13)		CDH80	1095	0	0	0	0	14	100	432	443	96	9	1	0	(13)
(14)	Wind	WSAvg	12.0	12.8	13.9	13.6	12.8	11.7	11.5	11.2	10.9	10.7	10.3	11.8	13.3	(14)
(15)	Precipitation	PrecAvg	29.6	4.4	4.0	3.2	2.4	1.2	0.3	0.0	0.9	3.0	4.9	5.7	(15)	
(16)		PrecMax	74.3	20.0	19.1	12.3	6.8	4.6	2.1	0.3	0.4	5.1	8.0	18.9	25.7	(16)
(17)		PrecMin	13.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	12.2	4.3	3.4	2.6	1.7	1.1	0.4	0.1	0.1	1.2	2.4	4.1	5.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.6	69.1	73.8	75.5	82.3	86.3	93.2	91.2	86.0	80.6	74.7	68.4	(19)
(20)			MCWB	57.5	58.2	59.7	61.4	64.9	67.2	69.0	69.7	68.7	68.1	62.8	59.3	(20)
(21)		2%	DB	64.6	66.0	69.1	71.9	77.4	82.7	87.8	87.6	82.6	77.1	71.2	66.2	(21)
(22)			MCWB	57.0	56.9	57.7	59.9	63.6	67.0	68.5	68.7	67.7	66.3	62.2	59.2	(22)
(23)		5%	DB	62.9	63.9	66.4	69.6	74.4	80.1	84.3	84.5	80.2	74.7	68.7	64.5	(23)
(24)			MCWB	56.6	56.6	57.2	59.3	62.5	66.2	68.1	68.5	68.0	65.9	61.8	58.6	(24)
(25)		10%	DB	61.4	62.2	64.6	67.7	72.0	77.4	81.7	82.1	77.9	72.9	67.1	62.9	(25)
(26)			MCWB	56.0	56.2	57.0	58.8	62.0	66.2	68.1	69.3	68.6	65.8	61.5	58.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.0	60.9	62.0	64.7	68.3	71.9	75.1	75.8	74.3	71.0	66.6	62.9	(27)
(28)			MADB	64.0	64.9	67.7	70.9	75.7	77.8	79.6	80.0	78.0	75.0	69.9	65.9	(28)
(29)		2%	WB	59.2	59.3	60.5	62.4	66.4	70.2	73.6	74.6	72.9	69.2	65.0	61.2	(29)
(30)			MADB	62.4	63.1	65.2	67.9	72.8	76.4	78.6	79.0	76.7	73.0	68.4	64.0	(30)
(31)		5%	WB	58.0	58.3	59.4	61.2	65.0	69.2	72.5	73.7	71.9	68.3	63.7	60.0	(31)
(32)			MADB	61.4	62.0	63.8	66.4	70.9	75.4	77.8	78.2	75.9	72.1	67.4	63.0	(32)
(33)		10%	WB	57.1	57.2	58.5	60.3	63.9	68.3	71.6	72.9	70.8	67.4	62.5	59.1	(33)
(34)			MADB	60.6	60.9	62.9	65.2	69.4	74.3	77.1	77.5	75.0	71.1	66.3	62.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	8.8	9.2	10.0	10.4	11.4	12.4	11.7	10.0	8.5	8.4	7.7	(35)
(36)			MADB	10.5	11.5	12.7	13.1	14.4	15.5	17.2	16.6	13.7	11.4	10.1	9.4	(36)
(37)			MCWBR	6.9	6.6	6.6	6.4	6.5	6.5	7.2	7.1	5.8	5.4	5.8	5.9	(37)
(38)		5% WB	MADB	8.4	9.2	10.0	10.1	11.4	11.7	12.0	10.4	8.6	8.3	7.9	7.1	(38)
(39)			MCWBR	6.5	6.3	6.4	6.3	6.3	6.1	6.8	6.1	4.8	4.9	5.3	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.336	0.361	0.402	0.400	0.416	0.437	0.438	0.450	0.426	0.397	0.360	0.341	(40)	
(41)		taud	2.511	2.439	2.308	2.340	2.323	2.271	2.263	2.235	2.333	2.416	2.488	2.511	(41)	
(42)		Ebn at Noon	272	278	274	280	276	269	268	262	263	261	260	262	(42)	
(43)		Edh at Noon	26	31	39	39	41	43	43	43	37	32	26	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	842	1087	1442	1800	2147	2455	2486	2190	1694	1228	895	740	(44)	
(45)		RadStd	80	134	156	130	154	87	54	78	95	113	76	75	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112